

Polina Petrov, PhD

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Education

Vanderbilt University , Nashville, TN	Aug 2026
Ph.D. in Astrophysics	
Thesis: "Seeing Double: Preparing for Multi-Messenger Detections of Supermassive Black Hole Binary Systems"	
Advisor: Dr. Stephen R. Taylor	
Carnegie Mellon University , Pittsburgh, PA	May 2021
B.S. in Physics, Astrophysics Concentration	
Mellon College of Science Honors and University Honors	

Honors & Awards

2026	Travel Award, APS DAP	\$500
2025	McMinn Summer Research Award, Physics and Astronomy, Vanderbilt University	\$1000
2024	Graduate School Conference Travel Grant, Vanderbilt University	\$1000
2024 - 2026	Harold Stirling Vanderbilt Fellowship, Vanderbilt University	
2023	Travel Award, APS DGRAV	\$1000
2023 - 2026	FINESST Grant #80NSSC23K1442, NASA	\$150,000
2021 - 2024	University Graduate Fellowship, Vanderbilt University	

Teaching Experience

Mentoring

Tejas Mahesh, Vanderbilt University undergraduate	June 2025 - Present
Celine Mang, Vanderbilt University undergraduate	Spring 2024

Workshops

Individual Supermassive Black Hole Binaries and Continuous Waves, NANOGrav Student Workshop, Ann Arbor, MI	Oct 2024
Continuous Waves, VIPER PTA Summer School, Vanderbilt University	July 2024

Graduate Teaching Assistant, Vanderbilt University

ASTR 1010L: Introductory Nighttime Astronomy Laboratory	2021 - 2023
ASTR 1020L: Introductory Daytime Astronomy Laboratory	Fall 2022

Publications

First Author

P. Petrov, L. Schult, S. R. Taylor, N. Pol, N. Laal, M. Charisi, C. P. Ma, "Expectations for the first supermassive black-hole binary resolved by PTAs II: Milestones for binary characterization", PRD, 114, 2 (2026).

DOI: [10.1103/zip2x-r4mp](https://doi.org/10.1103/zip2x-r4mp)

P. Petrov, S. R. Taylor, M. Charisi, C. P. Ma, "Identifying the Host Galaxies of Supermassive Black Hole Binaries"

Found by Pulsar Timing Arrays", ApJ, 976, 129 (2024).

DOI: [10.3847/1538-4357/ad7b14](https://doi.org/10.3847/1538-4357/ad7b14)

P. Petrov, L. P. Singer, M. W. Coughlin, [and 7 others], "*Data-driven expectations for electro-magnetic counterpart searches based on LIGO/Virgo public alerts*", ApJ, 924, 54 (2022).

DOI: [10.3847/1538-4357/ac366d](https://doi.org/10.3847/1538-4357/ac366d)

Short Author

L. Schult, **P. Petrov**, S. R. Taylor, N. Pol, N. Laal, M. Charisi, C. P. Ma, "*Expectations for the first supermassive black-hole binary resolved by PTAs I: Model efficacy*" PRD, 114, 2 (2026).

DOI: [10.1103/vncm-5j2n](https://doi.org/10.1103/vncm-5j2n)

L. Schult, **P. Petrov**, S. R. Taylor, N. Pol, N. Laal, M. Charisi, C. P. Ma, "*Expectations for the first supermassive black-hole binary resolved by PTAs I: Model efficacy*" PRD, 114, 2 (2026).

DOI: [10.1103/vncm-5j2n](https://doi.org/10.1103/vncm-5j2n)

N. Veronesi, M. Charisi, **P. Petrov**, S. R. Taylor, J. Runnoe, D. J. D'Orazio, J. Pilawa, C. P. Ma, "*Prospects of resolving and localising individual supermassive black hole binaries with pulsar timing arrays: the host ranking challenge*", (2026).

arXiv: [2606.00218](https://arxiv.org/abs/2606.00218)

M. Charisi, S. R. Taylor, J. Runnoe, C. A. Witt, and **P. Petrov**, "*Multi-messenger Analysis of Supermassive Black Hole Binaries: The Joint-likelihood Approach*", MNRAS, 548, 3 (2026).

DOI: [10.1093/mnras/stag477](https://doi.org/10.1093/mnras/stag477)

S. Burke-Spolaor, T. Bogdanović, D. J. D'Orazio, [and 9 others, **including P. Petrov**], "*The Era of Binary Supermassive Black Holes: Coordination of Nanohertz-Frequency Gravitational-Wave Follow-up*", BAAS, 58, 1 (2026).

DOI: [10.3847/25c2cfcb.9aea5fa1](https://doi.org/10.3847/25c2cfcb.9aea5fa1)

B. Bécsy, N. J. Cornish, **P. Petrov**, X. Siemens, S. R. Taylor, S. J. Vigeland, and C. A. Witt, "*Towards robust gravitational wave detections from individual supermassive black hole binaries*", CQG, 42, 17 (2025).

DOI: [10.1088/1361-6382/adfd36](https://doi.org/10.1088/1361-6382/adfd36)

Collaboration Publications

The NANOGrav Collaboration, "*Gravitational-wave Measurement of the $M_{\text{BH}} - M_{\text{bulge}}$ Intrinsic Scatter at High Redshift*", ApJ, 1006, 63 (2026).

DOI: [10.3847/1538-4357/ae801e](https://doi.org/10.3847/1538-4357/ae801e)

The NANOGrav Collaboration, "*The NANOGrav 15 yr Data Set: Impacts of Customized Chromatic Noise Models on Gravitational Wave Analyses*", (2026).

arXiv: [2606.28554](https://arxiv.org/abs/2606.28554)

The NANOGrav Collaboration, "*The NANOGrav 15 yr Data Set: Piecewise Power-Law Reconstruction of the Gravitational-Wave Background*", ApJL, 1004, L40 (2026).

DOI: [10.3847/2041-8213/ae7086](https://doi.org/10.3847/2041-8213/ae7086)

The NANOGrav Collaboration, "*Inference on inner galaxy structure via gravitational waves from supermassive binaries*", Nat. Astron., 10, 554–563 (2026).

DOI: [10.1038/s41550-026-02782-0](https://doi.org/10.1038/s41550-026-02782-0)

The NANOGrav Collaboration, "*The NANOGrav 15 yr Data Set: Targeted Searches for Supermassive Black Hole Binaries*", ApJL, 998, L11 (2026).

DOI: [10.3847/2041-8213/ae3719](https://doi.org/10.3847/2041-8213/ae3719)

The NANOGrav Collaboration, "*Inferring M_{bh} - M_{bulge} Evolution from the Gravitational Wave Background*", ApJ, 997, 188 (2026).

DOI: [10.3847/1538-4357/ae2480](https://doi.org/10.3847/1538-4357/ae2480)

The NANOGrav Collaboration, "*The NANOGrav 15 yr Data Set: Search for Gravitational-wave Memory*", ApJ, 987, 5 (2025).

DOI: [10.3847/1538-4357/add874](https://doi.org/10.3847/1538-4357/add874)

The NANOGrav Collaboration, "*The NANOGrav 15 yr Data Set: Harmonic Analysis of the Pulsar Angular Correlations*", ApJ, 985, 99 (2025).

DOI: [10.3847/1538-4357/adc997](https://doi.org/10.3847/1538-4357/adc997)

The NANOGrav Collaboration, "*Erratum: 'The NANOGrav 15 yr Data Set: Search for Signals from New Physics' (2023, ApJL 951 L11)*", ApJL, 971, L27 (2024).

DOI: [10.3847/2041-8213/ad68fc](https://doi.org/10.3847/2041-8213/ad68fc)

The IPTA Collaboration, "*Comparing Recent Pulsar Timing Array Results on the Nanohertz Stochastic Gravitational-wave Background*", ApJ, 966, 105 (2024).

DOI: [10.3847/1538-4357/ad36be](https://doi.org/10.3847/1538-4357/ad36be)

The NANOGrav Collaboration, "*The NANOGrav 15 yr Data Set: Constraints on Supermassive Black Hole Binaries from the Gravitational Wave Background*", ApJL, 952, L37 (2023).

DOI: [10.3847/2041-8213/ace18b](https://doi.org/10.3847/2041-8213/ace18b)

The NANOGrav Collaboration, "*The NANOGrav 15 yr Data Set: Search for Signals from New Physics*", ApJL, 951, L11 (2023).

DOI: [10.3847/2041-8213/acdc91](https://doi.org/10.3847/2041-8213/acdc91)

The NANOGrav Collaboration [including **P. Petrov** in analysis team], "*The NANOGrav 15 yr Data Set: Bayesian Limits on Gravitational Waves from Individual Supermassive Black Hole Binaries*", ApJL, 951, L50 (2023).

DOI: [10.3847/2041-8213/ace18a](https://doi.org/10.3847/2041-8213/ace18a)

The NANOGrav Collaboration [including **P. Petrov** in analysis team], "*The NANOGrav 15 yr Data Set: Evidence for a Gravitational-Wave Background*", ApJL, 951, L8 (2023).

DOI: [10.3847/2041-8213/acdac6](https://doi.org/10.3847/2041-8213/acdac6)

Presentations

Needle in a Haystack: Ranking Supermassive Black Hole Binary Hosts

APS Global Summit, Denver, CO

Mar 2026

Detection Milestones for Individual Binaries Detected by PTAs

IPTA Science Meeting, California Institute of Technology, Pasadena, CA

June 2025

Identifying Host Galaxies of Supermassive Black Hole Binaries Found by PTAs

The Era of Binary Supermassive Black Holes Meeting, Aspen, CO

Feb 2025

Astronomy Department Lunch Talk, University of California, Berkeley, CA

Dec 2024

IPTA Science Meeting, Sexten, Italy

June 2024

Optimizing Host Galaxy Identification of Individual Supermassive Black Hole Binaries

NANOGrav Spring Meeting, Virtual

Mar 2024

AAS 243, New Orleans, LA

Jan 2024

NANOGrav Fall Meeting, University of British Columbia, Vancouver, BC

Oct 2023

Mapping the Host Galaxies of Nanohertz Gravitational Wave Sources

IPTA Science Meeting, Virtual

June 2023

APS April Meeting, Minneapolis, MN

Apr 2023

NANOGrav Fall Meeting, University of Wisconsin-Milwaukee, Milwaukee, WI

Oct 2022

GMT Meeting: Black Holes at All Scales, Sedona, AZ (Poster)

Sept 2022

VIPER PTA Summer School, Nashville, TN

July 2022

Astronomy Journal Research Club, Vanderbilt University

Multi-messenger Analysis of Supermassive Black Hole Binaries: The Joint-likelihood Approach	Oct 2025
Identifying Host Galaxies of Supermassive Black Hole Binaries Found by PTAs	Feb 2025
The NANOGrav 15-year Data Set: Bayesian Limits on Gravitational Waves from Individual Supermassive Black Hole Binaries	Jan 2024
The NANOGrav 12.5-year Data Set: Bayesian Limits on Gravitational Waves from Individual Supermassive Black Hole Binaries	Jan 2023

Service & Outreach

Astronomy on Tap organizer and social media coordinator, Nashville, TN	2023 - 2026
Astronomy on Tap talk, Nashville, TN	April 2026
Adventure Science Center Way Late Play Date, Vanderbilt Astronomy Outreach	May 2025
NANOGrav Student Workshop Organizing Committee, Ann Arbor, MI	Oct 2024
VIPER PTA Summer School organizer, Vanderbilt University	July 2024, 2022
Radio interview on WPLN News, Nashville, TN	July 2023
Adventure Science Center Astronomy Day, Vanderbilt Astronomy Outreach	April 2023
Physics Concepts mentor, Carnegie Mellon University	2018
Buhl Planetarium volunteer, Pittsburgh, PA	2016 - 2018

Skills

Programming — Python, Bash, High-Performance Computing (SLURM)
Scientific Software — enterprise-pulsar, libstempo, QuickCW, ligo.skymap
Other — LaTeX, HTML, Github Pages, Microsoft Office, Google Docs, Adobe Illustrator

Professional & Academic Affiliations

Full Member, North American Nanohertz Observatory for Gravitational waves (NANOGrav)
 Member, International Pulsar Timing Array (IPTA)
 Member, American Physical Society
 Member, American Astronomical Society